

Akarshan Rasyal

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Summary

MSc Data Science graduate specialising in Python, SQL, Machine Learning, Statistics, NLP and AI. Skilled at turning messy transactional data into reliable pipelines, validated metrics, predictive models and decision-support tools. Quick learner and strong problem solver who enjoys tackling unfamiliar problems. Earned HackerRank SQL (Advanced) Certificate - [Link](#).

Key Skills

Programming & Data: Python, SQL, Pandas, NumPy

Machine Learning & Statistics: Probability, Regression, Classification, Clustering, Time-Series Forecasting, Hyperparameter Tuning, SHAP, Model Evaluation, CNN, RNN, Experiment Design, Causal Inference, A/B Testing, Gradient Boosting, Ranking, Pytorch, Deep learning

Data Engineering & Cloud: SQL, AWS, Snowflake, Databricks, Docker, GitHub, CI/CD, Data Pipelines, Data Modelling.

AI & NLP: LLMs, RAG, Prompt Engineering, LangChain, Embeddings, Vector Search, Claude, CoPilot

Business Analytics: Customer Analytics, Pricing, Forecasting, Segmentation, Anomaly/Drift Detection, Power BI

Soft Skills: Reasoning, Problem Solving, Responsible, Clear Communication, English, Team Player

Projects

Retail Customer Intelligence & Pricing Analytics Platform ([GitHub Link](#)) | ([LIVE LINK](#)) | ([Demo Video](#))

Python | SQL | Scikit-learn | LightGBM | XGBoost | Econometrics | Streamlit | Plotly | Docker

- Designed and developed scalable end-to-end data pipelines and analytics workflows to process and validate 2.59M retail transactions across 92K+ products, enabling reliable data preparation, metric definitions and feature engineering spanning customer intelligence, revenue modelling, demand forecasting and pricing analytics.
- Built and evaluated machine learning models achieving 92.40% churn classification accuracy, 98.34% demand forecasting R^2 , and +22.53% Forecast Value Added over a naive baseline.
- Developed 25 category-level price elasticity models, achieving 99.18% combined portfolio R^2 , 94.45% WAPE accuracy, and 19/25 statistically significant models ($p < 0.05$).
- Delivered an interactive Streamlit decision-support application integrating customer predictions, demand forecasting, pricing simulations, model governance and statistical validation, with reproducible deployment using Docker.

AI Deepfake & Synthetic Speech Anti-Spoofing Detection ([GitHub Link](#)) | ([Demo Video](#))

Python | PyTorch | TorchAudio | Librosa | Scikit-learn | XGBoost

- Engineered an end-to-end audio deepfake detection pipeline across 10,000 audio samples, 93 speakers, 100+ attack algorithms and 9 lossy transmission codecs, using strict zero-speaker-leakage train, validation and test partitions.
- Developed and benchmarked multi-domain acoustic models using 425+ engineered features, 3-channel LFCC ($LFCC + \Delta + \Delta^2$) with CMVN, high-resolution STFT representations, LCNN-29, SE-ResNet, and ensemble learning.
- Improved test accuracy from 76.40% to 90.73%, achieved 96.56% ROC-AUC, and reduced Equal Error Rate (EER) by 61.7%, from 24.00% to 9.20%, on completely unseen speakers.
- Performed error slicing across vocoder architectures and lossy audio codecs and calibrated decision thresholds to reduce false-negative deepfake detections by 70.1%.

AI Video Intelligence Platform — Prediction & Recommendation ([GitHub Link](#)) | ([AWS LIVE LINK](#)) | ([Demo Video](#))

Python, React, LLMs, RAG, LangChain, Embeddings, Vector Search, Speech-to-Text, Docker, AWS, API, Rest, Json

- Built an end-to-end AI platform that transforms educational videos into searchable, interactive knowledge using speech-to-text transcription, semantic search and a RAG-based conversational AI system.
- Developed an Extra Trees predictive model to forecast learners' next quiz scores, achieving $R^2 = 0.861$ and 3.13% MAE.
- Designed a personalised topic-recommendation engine that identifies learner knowledge gaps and recommends relevant videos to support targeted study.
- Integrated a RAG chatbot with timestamp-aware retrieval, enabling learners to ask questions and navigate directly to the relevant point in a video through clickable timestamps.

Enterprise SQL Data Warehouse & BI Analytics Platform ([GitHub Link](#))

Microsoft SQL Server, Stored Procedures, Medallion Architecture, ETL, Star Schema, Power BI Desktop, DAX

- Architected an end-to-end SQL Data Warehouse using Microsoft SQL Server and Presented using Power BI.

Education

MSc Data Science | **Northumbria University** | Jan 2025 – June 2026 | Newcastle, UK (64%)

Computer Science Engineering | **DBATU** | Jan 2020 – Jul 2024 | Maharashtra, India (6.90 CGPA)